Date: 24/02/2026

Government Engineering College, Rajkot

General Department
Mid exam syllabus

Subject: Discrete Mathematics and Graph Theory (BE04000261)

Branch: CE and AIDS **Semester:4**

Unit	CO	Content
Unit 1	1	**Propositional Logic**: Definition, statements & notation, truth values, connectives, statement formulas & truth tables, well-formed formulas, tautologies, equivalence of formulas, duality law, tautological implications, examples. **Predicate Logic**: Definition of predicates, statement functions, variables, quantifiers, predicate formulas, free & bound variables, the universe of discourse, examples, valid formulas & equivalences, examples
Unit 2	2	**Set Theory**: Basic concepts of set theory like inclusion, complement, intersection, union, Cartesian product, power set. **Counting and Combinatorics:** Basic counting principles, permutations and combinations, binomial coefficients and the binomial theorem, Pigeonhole principle, inclusion-exclusion principle. Recurrence relations: Definitions and simple examples only.
Unit 5	5	**Graphs:** Introduction, definition, examples; Nodes, edges, adjacent nodes,directed and undirected edge, directed graph, undirected graph, examples;Initiating and terminating nodes, loop (sling), distinct edges, parallel edges,multi-graph, simple graph, weighted graphs, examples; Isolated nodes, nullgraph, isomorphic graphs, examples; Degree, In-degree, out-degree, total degree of a node, examples; Subgraphs: definition, examples; Converse(reversal or directional dual) of a digraph, examples; Path: Definition, paths of a given graph, length of path, examples; Simple path (edge simple), elementary path (node simple), examples; Cycle (circuit), elementary cycle, examples; **Reachability:** Definition, geodesic, distance, examples; Properties ofreachability, the triangle inequality; Reachable set of a given node, examples; Node base, examples Connectedness: Definition, weakly connected, strongly connected, unilaterally connected, examples; Strong, weak, and unilateral components of a graph, examples; Applications to represent resource allocation status of an operating system, and detection and correction of deadlocks; Matrix representation of graph: Definition, adjacency matrix, Boolean (or bit) matrix, examples; Determine number of paths of length n through adjacency matrix, examples; Path (Reachability) matrix of a graph, examples; Warshall's algorithm to produce path matrix, Flowchart. **Trees:** Definition, branch nodes, leaf (terminal) nodes, root, examples; Different representations of a tree, examples; Binary tree, m-ary tree, full (or complete) binary tree, examples; Converting any m-ary tree to a binary tree, examples; Representation of a binary tree: Linked-list; Tree traversal: Pre-order, in-order, post-order traversal, examples; Algorithms; Applications of List structures and graphs

Subject Coordinator
Prof. N. D. Abhangi